Endometriosis AI Analysis Report

Advanced Multi-Modal Federated Prediction

1. Patient Clinical Metrics

Age	BMI	Pelvic Pain	Dysmenorrhea
28	22.1	8/10	9/10

Dyspareunia	Family Hist.	CA-125	Estradiol
Yes	Yes	78.5 U/mL	285.0 pg/mL

Progest.
4.2 ng/mL

2. AI Diagnostic Prognosis

Confidence Score: 0.5% (+/- 4.0%)

Estimated Stage: Stage II (Mild)

Future Risk Trajectory (Years 1, 3, 5):

1-Year: 43.8%, 3-Year: 48.3%, 5-Year: 51.1%

Generative Clinical Synthesis:

DISCLAIMER: AI research tool not a clinical diagnosis.

Primary Assessment: 0.5% (+/-4.0%) probability of active endometriosis. Estimated classification: Stage II (Mild).

Neural Routing: 39.1% of computation routed to the Adenomyosis/Uterine Expert Sub-Network.

Biomarker Highlights:

- CA-125: 78.5 U/mL elevated (threshold 35 U/mL). Differential diagnosis required.
- Estradiol: 285.0 pg/mL hyperestrogenism may drive ectopic proliferation.
- Pelvic Pain Score 8/10 indicates significant nociceptive involvement.

Recommendation: High-resolution transvaginal ultrasound and pelvic MRI for structural assessment.

Personal Actionable Plan:

Personal Actionable Health Plan (AI-generated for educational purposes only):

1. Monitoring: Risk profile is stable. Maintain routine gynaecological checkups.
2. Pain Management: Pain score is elevated discuss NSAIDs or pelvic floor physiotherapy.
3. Hormonal Balance: Omega-3-rich anti-inflammatory diet may help regulate excess oestrogen.

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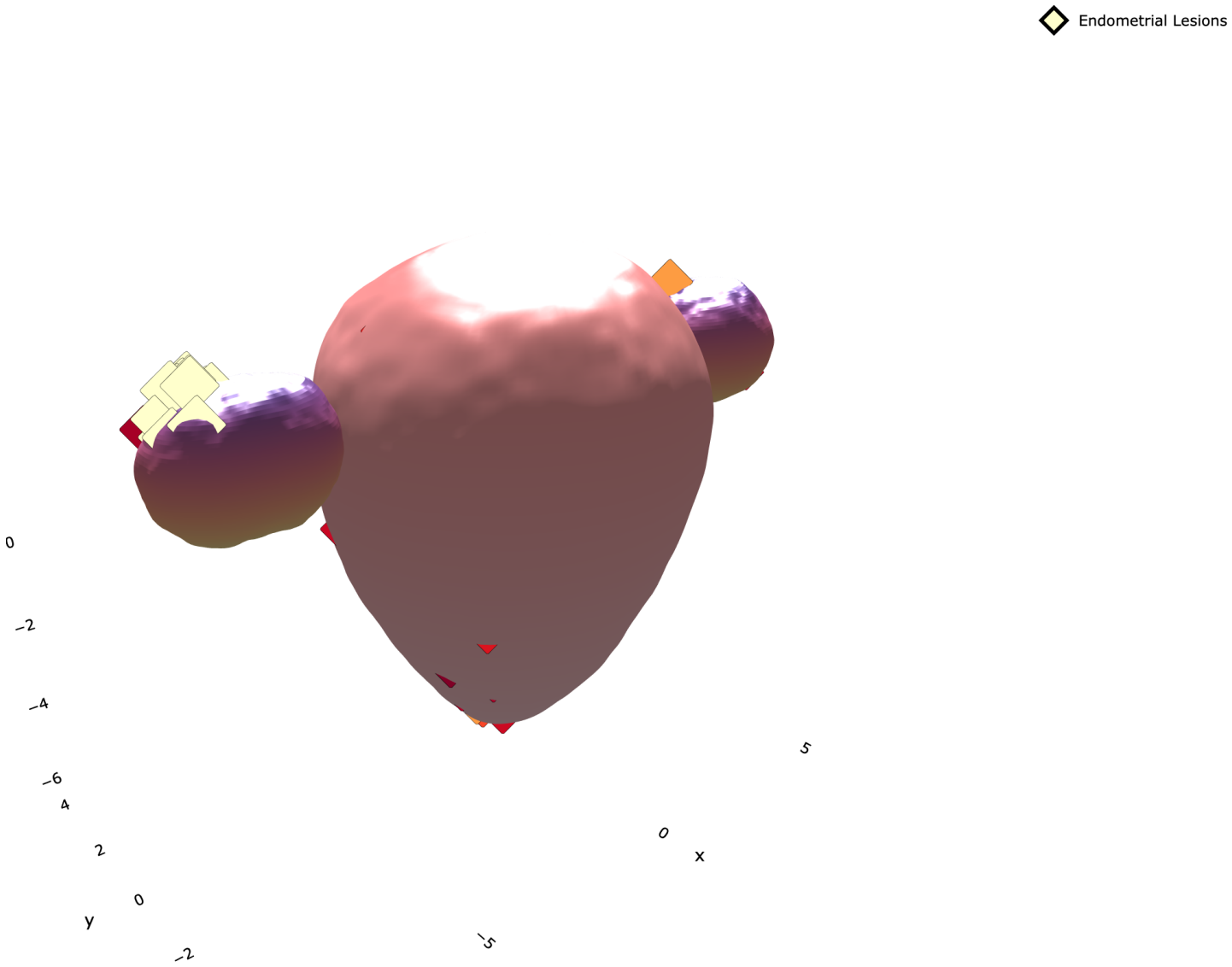
3. Physics-Informed Digital Twin (3D Render)

High-resolution rendering of the generated patient organ geometry, mapping lesions, endometriomas, and simulated adhesions based on the PINN predictions.



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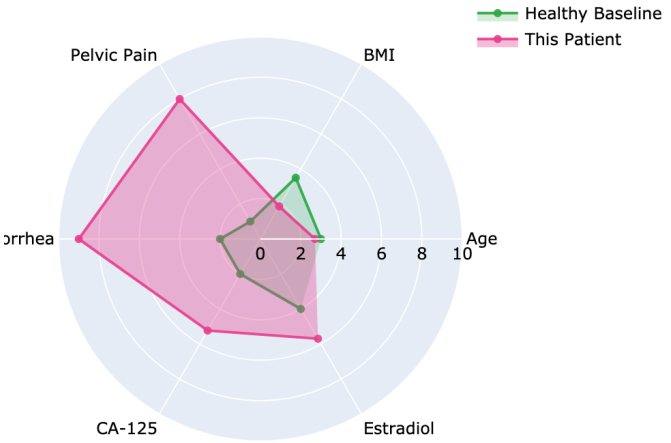
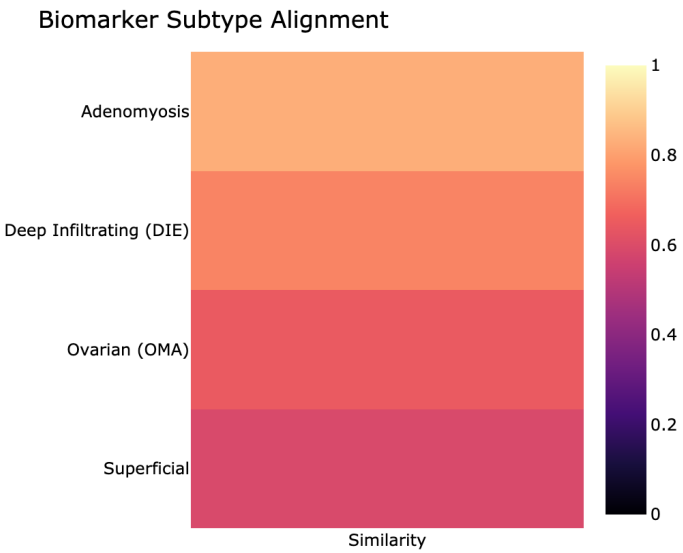
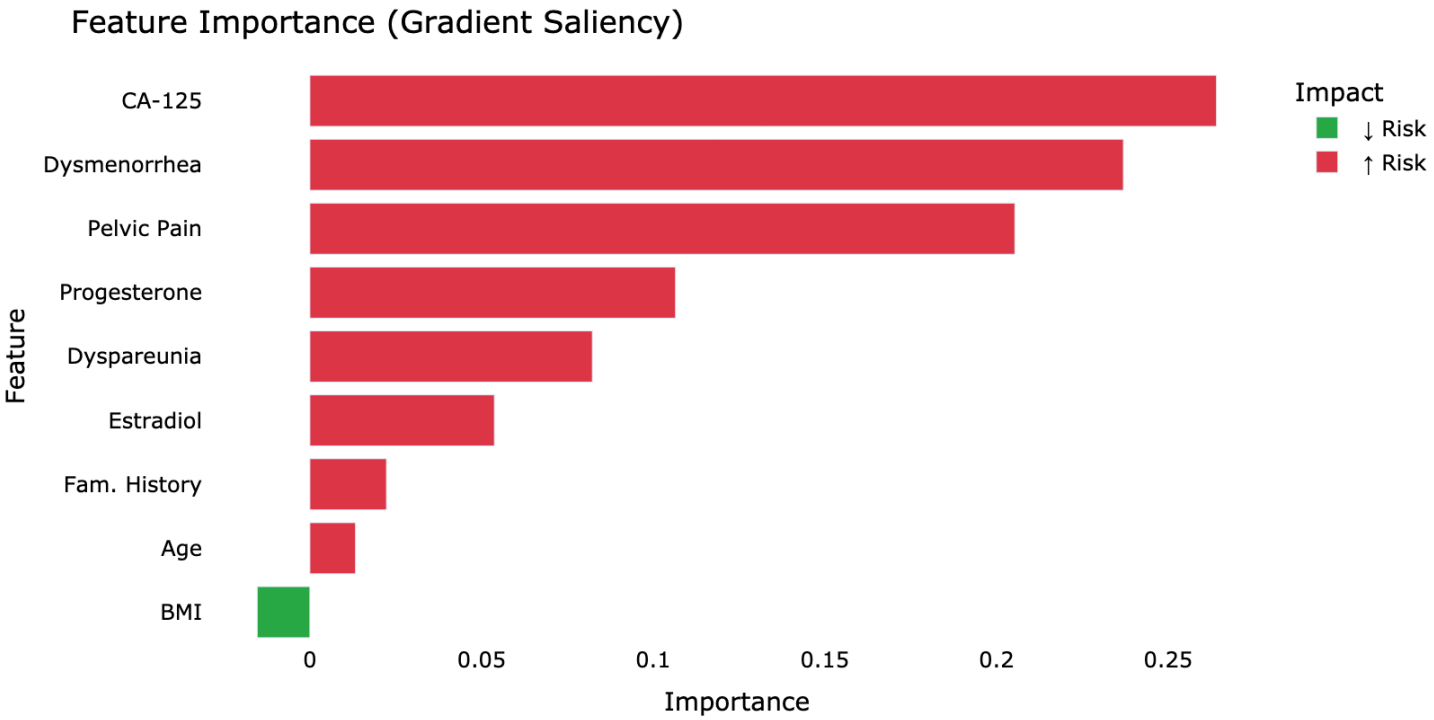
4. Advanced Analytics & Explainable AI





Feature Importance (SHAP/XAI Explainer):



XAI Insight: Risk score 0.5%. Primary drivers: Dysmenorrhea (9.0), CA-125 (78.5).